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11. Agricultural role

- It is used to eradicate weeds. Example: 2,4-D and 2,4,5-T.
- Synthetic auxins are used in the formation of seedless fruits (Parthenocarpic fruit).
- It is used to break the dormancy in seeds.
- Induce flowering in Pineapple by NAA & 2,4-D.
- Increase the number of female flowers and fruits in cucurbits.

15.2.2 Gibberellins

1. Discovery

The effect of gibberellins had been known in Japan since early 1800 where certain rice plants were found to suffer from '**Bakanae**' or foolish seedling disease. This disease was found by **Kurosawa** (1926) to be caused by a fungus *Gibberella fujikuroi*. The active substance was separated from fungus and named as gibberellin by **Yabuta** (1935). These are more than 100 gibberellins reported from both fungi and higher plants. They are noted as GA_1, GA_2, GA_3 and so on. GA_3 is the first discovered gibberellin. In 1938, **Yabuta** and **Sumiki** isolated gibberellin in crystalline form. In1955, **Brain** *et al.*, gave the name **gibberellic acid**. In 1961, Cross *et al.*, established its structure.

2. Occurrence

The major site of gibberellin production in plants is parts like embryo, roots and young leaves near the tip. Immature seeds are rich in gibberellins.

3. Precursors

The gibberellins are chemically related to terpenoids (natural rubber, carotenoids and steroids) formed by 5-C precursor, an Isoprenoid unit called Iso Pentenyl Pyrophosphate (IPP) through a number of intermediates. The primary precursor is acetate.

4. Chemical structure

All gibberellins have gibbane ring structure.

5. Transport in plants

The transport of gibberellins in plants is non-polar. Gibberellins are translocated through phloem and also occur in xylem due to lateral movement between vascular bundles.

6. Bioassay (Dwarf Pea assay)

Seeds of dwarf pea are allowed to germinate till the formation of the coleoptile. GA solution is applied to some seedlings. Others are kept under control. Epicotyle length is measured and as such, GA stimulating epicotyle growth can be seen.

7. Physiological Effects

- It produces extraordinary elongation of stem caused by cell division and cell elongation.
- Rosette plants (genetic dwarfism) plants exhibit excessive internodal growth when they are treated with gibberellins. This sudden elongation of stem followed by flowering is called **bolting** (Figure 15.13).
- Gibberellin breaks dormancy in potato tubers.
- Many biennials usually flower during second year of their growth. For flowering to take place, these plants should be exposed to cold season. Such plants could be made to flower without